

Practical - 1

Problem Statement: Write a program to Compute Similarity between two text documents.

```
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
import numpy as np
import nltk

nltk.download("punkt")
nltk.download("stopwords")
nltk.download('punkt_tab')

[nltk_data] Downloading package punkt to /home/nonroot/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data] /home/nonroot/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data] /home/nonroot/nltk_data...
[nltk_data] Package punkt_tab is already up-to-date!

True

def print_file_content(file):
    with open(file, 'r') as f:
        content = f.read()
        print(content)

print("Contents of text1.txt:")
print_file_content("text1.txt")
print("Contents of text2.txt:")
print_file_content("text2.txt")

Contents of text1.txt:
This is a sample document. It contains text for testing the
similarity.

Contents of text2.txt:
This document is a sample. It includes text to test the similarity.

def process(file):
    # Read the file
    raw = open(file).read()
```

```

# Tokenize the raw text
tokens = word_tokenize(raw)
words = [w.lower() for w in tokens]

# Stem the tokens
porter = nltk.PorterStemmer()
stemmed_tokens = [porter.stem(t) for t in words]

# Removing stop words
stop_words = set(stopwords.words('english'))
filtered_tokens = [w for w in stemmed_tokens if w not in
stop_words]

# Count words
count = nltk.defaultdict(int)
for word in filtered_tokens:
    count[word] += 1

return count

def cos_sim(a, b):
    dot_product = np.dot(a, b)
    norm_a = np.linalg.norm(a)
    norm_b = np.linalg.norm(b)

    return dot_product / (norm_a * norm_b)

def getSimilarity(dict1, dict2):
    all_words_list = []

    for key in dict1:
        all_words_list.append(key)

    for key in dict2:
        all_words_list.append(key)

    all_words_list_size = len(all_words_list)
    v1 = np.zeros(all_words_list_size, dtype=int)
    v2 = np.zeros(all_words_list_size, dtype=int)

    for i, key in enumerate(all_words_list):
        v1[i] = dict1.get(key, 0)
        v2[i] = dict2.get(key, 0)

    return cos_sim(v1, v2)

if __name__ == '__main__':
    dict1 = process("text1.txt")
    dict2 = process("text2.txt")

```

```
print("Similarity between two text documents:",  
getSimilarity(dict1, dict2))
```

```
Similarity between two text documents: 0.9523809523809523
```
